MAT 320 Fall 2024

Aditya Pahuja

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Contents

These are notes for MAT 320 only (i.e. starting week 7).

7 Week 7: 10/08 and 10/10

The following two theorems are some of the most important in analysis over $\mathbb{R}$.

Theorem 7.1 (Weierstrass theorem). If $f : \mathbb{R} \to \mathbb{R}$ is a continuous function on the interval $[a, b]$, then it has a maximum and a minimum. More precisely, the set $\{f(x) : x \in [a, b]\}$ has a minimum and a maximum.

Theorem 7.2 (Intermediate value theorem). If $f : \mathbb{R} \to \mathbb{R}$ is continuous on an interval I (whether open, half-open, or closed) and $a < b$ are two elements of I, then, if y is between $f(a)$ and $f(b)$ (so either $f(a) < y < f(b)$ or $f(b) < y < f(a)$), there exists an $x \in (a, b)$ such that $f(x) = y$.

Exercise 7.3. Show that the intermediate value theorem equivalently says that $f(I)$ is also an interval.

Some toy applications of these theorems:

Theorem. Let $y \geq 0$ be real and m an integer. Then, there exists a real $x \geq 0$ such that $x^m = y$.

Note that there is no uniqueness assertion here!

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = x^m$. We check three cases.

- If $y = 0$ or $y = 1$, then the result is trivial, as we take $x = 0$, $x = 1$ respectively.
- If $0 < y < 1$, then $f(0) < y < f(1)$ implies that there exists an $x \in (0, 1)$ such that $f(x) = y$, by the intermediate value theorem.
- If $1 < y$, then $f(1) < y < f(y)$ implies that there exists an $x \in (1, y)$ such that $f(x) = y$, again by the intermediate value theorem.

These cases encompass all possible values of y, so we are done. □

Note in fact that we got an estimate on how large x must be for a given y, too! (A bad estimate, but still.)

Some more problems for the reader:

Problem 7.4

Show that if $f : [0, 1] \to [0, 1]$ is continuous then f has a fixed point; that is, there exists an $x \in [0, 1]$ such that $f(x) = x$.

Problem 7.5

On Saturday, a mountain climber climbs from the base of a mountain to its peak between 7 am and 10 pm. On Sunday, the mountain climber climbs down from the peak to the base, again between 7 am and 10 pm. Prove that at some point on Sunday, the mountain climber is at the same altitude that he was at exactly 24 hours ago.

Rather than just proving these theorems for real numbers, we will depart into the world of *metric spaces* to help us understand them in a much deeper context.

7.1 Metric spaces

Fundamentally, there is really only one thing that matters in the context of the basic ideas of convergence (which is what basically everything so far has been based on): the notion of a distance. In light of this, we momentarily strip away all the extra features of $\mathbb{R}$ to get the following.

Definition 7.6 (Metric space) — A **metric space** is a set X equipped with a **metric** ("distance function") $d\colon X \times X \to \mathbb{R}$ such that, for all x, y, $z \in X$,

- $d(x, y) \geq 0$, with equality if and only if $x = y$;
- $d(x, y) = d(y, x)$; and
- $d(x, y) \leq d(x, z) + d(z, x)$.

We refer to metric spaces as a pair (X, d), or, if d is clear from context, then we will just say X. If we just say "X is a metric space," then the corresponding metric will be called d. We also often refer to the elements of X as "points" to align with visual intuition.

Example 7.7

Some example metric spaces:

- Any subset of the real numbers is a metric space with metric $d(x, y) = |x - y|$.
- In general, n-dimensional space is a metric space with the Euclidean metric

  $$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

  or with the taxicab metric

  $$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + \cdots + |x_n - y_n|$$

  or with the Chebyshev metric

  $$d(\mathbf{x}, \mathbf{y}) = \max\{|x_1 - y_1|, \ldots, |x_n - y_n|\}.$$

- A (connected) graph can be turned into a metric space with distance given by the length of the shortest path between two points.
- We also have the trivial metric

  $$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y. \end{cases}$$

- The rationals can also be made a metric space with distance $d(a, b) = p^{-\nu_p(a-b)}$; this is called the p-adic metric. Here $\nu_p(n)$ is the largest power of p dividing n if n is a nonzero integer, ∞ if $n = 0$, and we extend to rationals by decreeing that $\nu_p(a/b) = \nu_p(a) - \nu_p(b)$ for all rationals; note that we define $p^{-\infty} = 0$. Actually, the rationals with this metric form an ultrametric space, meaning the stronger version of the triangle inequality

  $$d(x, y) \leq \max\{d(x, z), d(y, z)\}$$

  always holds (this is not too hard to prove if you know some number theory).

Our job now is to carry over our basic definitions from $\mathbb{R}$ to general metric spaces. Recalling that $|x - y|$ just means "the distance between x and y" in $\mathbb{R}$, this is just a matter of copying the definitions and changing absolute values into distances. For example:

Definition 7.8 (Convergence) — Let (X, d) be a metric space and let (x_n) be a sequence of points in X. Then, we say that (x_n) converges to a point $\ell \in X$ if

$$\text{for all } \varepsilon > 0, \quad d(x_n, \ell) < \varepsilon \text{ eventually.}$$

Similarly, we can define what it means for a function to be continuous; note that since we compare distances both in the domain ("$|x - c| < \delta$") and in the codomain ("$|f(x) - f(x)| < \varepsilon$"), both the domain and codomain need to be metric spaces (but not necessarily the same one!).

Definition 7.9 (Continuity) — A function $f\colon X \to Y$ between metric spaces (X, d_X) and (Y, d_Y) is continuous at a point $c \in X$ if

$$\text{for all } \varepsilon > 0, \text{ there exists } \delta > 0 \text{ such that } d_X(x, c) < \varepsilon \implies d_Y(f(x), f(c)) < \varepsilon.$$

7.2 Some topology

At this point, you might have made an observation: the idiom $d(a, x) < \varepsilon$ keeps popping up, since we frequently need to refer to the idea of "x is less than ε away from a." To make notation less clunky, we define:

Definition 7.10 — In a metric space (X, d), the **open ball** centered at $x \in X$ with radius $\varepsilon > 0$ is the set of points

$$B_X(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\}.$$

The **open unit ball** is then just $B_X(x, 1)$. We sometimes omit the subscript (simply writing $B(x, \varepsilon)$) if there is no risk of confusion about the ambient metric space.

The adjective "open" here refers to the idea that this set doesn't have its natural "boundary" of points whose distance from x is exactly ε — the open balls in $\mathbb{R}^n$ should give you a clearer idea of what this means. We will see a slightly more specific example of what this means in a moment.

Exercise 7.11. Pick your three favorite metric spaces. What do open balls look like in those spaces?

Exercise 7.12. Rewrite Definitions 7.9 and 7.8 with the notation of open balls.

The kicker is that we very much care about sets $U \subseteq X$ such that, for every $x \in U$, some open ball $B(x, \varepsilon)$ with $\varepsilon > 0$ is contained in U, since these are contexts in which we can freely talk about convergence, continuity, and so on (we only require existence of an ε because we only ever care about "arbitrarily small ε" In fact, such U are so important that we give them a name:

Definition 7.13 (Open and closed sets) — Let (X, d) be a metric space as usual. A subset $U \subseteq X$ is said to be **open** (in X, with respect to the metric d — both of these descriptors are dropped if we have enough context) if, for every $x \in U$, there exists an ε such that $B(x, \varepsilon)$ is a subset of U.

The set $E \subseteq X$ is **closed** if its complement $X \setminus E$ is open.

Exercise 7.14. Show that every open ball is open in its respective metric space, this earning open balls their name.

Exercise 7.15. Let (X, d) be a metric space. Prove the following facts about open sets:

(i) $\varnothing$ and X are open in X.

(ii) The union of any (possibly infinite) collection of open sets is also an open set.

(iii) The intersection of finitely many open sets is also an open set. Show that finiteness is a necessary assumption here by finding an infinite collection of open sets in $\mathbb{R}$ whose intersection is not open in $\mathbb{R}$.

What are the analogous statements for closed sets? Write them down and prove them, too.

Closed sets are in fact aptly named too:

Theorem 7.16 (Alternate definition of closed set)**.** Let $E \subseteq X$. A point $x \in X$ is a **limit point** of E if there exists a sequence (x_n) of points in E whose limit is x. Then, E is closed if and only if it contains all of its limit points.

proof

Proof. □

You can think of this as "closed sets do contain their boundaries."

Exercise 7.17. Show that finite sets are closed in every metric space. Then, find a metric space in which all finite sets are open, and find a metric space in which no finite sets are open.

The following theorem justifies the above definitions by allowing us to think about continuity much more globally rather than pointwise (in fact, this will be the exact definition of continuity in general topological spaces):

Theorem 7.18 (Continuity, topologically)**.** A function $f : X \to Y$ between metric spaces is continuous if and only if, for each open $U \subseteq Y$, the set $f^{-1}(U) \subseteq X$ is also open. This theorem also holds if the word "open" is replaced with "closed."

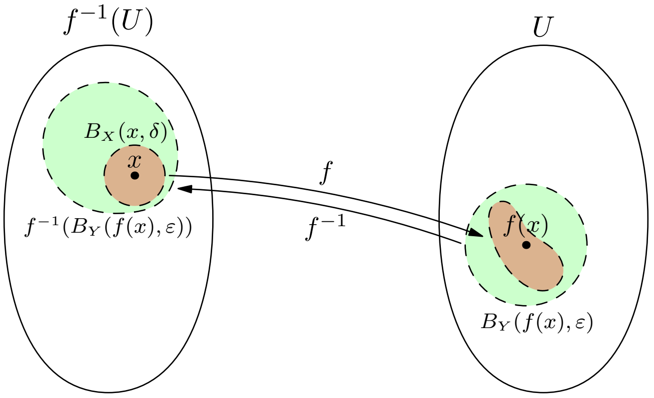
The proof of this is given in Week 8, and a formal writeup was required for homework 7. You should try to draw a picture and prove the theorem yourself before reading on; being able to prove it yourself means you can be confident in your understanding of the preceding content.

8 Week 8: 10/17

From the end of last week's lecture:

Theorem 8.1 (Continuity, topologically)**.** Let $f : X \to Y$ be a function between two metric spaces. Prove that f is continuous if and only if, for every open subset $U \subseteq Y$, the pre-image $f^{-1}(U)$ is open in X.

Proof. Here's a picture:



First, suppose f is continuous. For an open $U \subseteq Y$, let $x \in f^{-1}(U)$ and let $B_Y(f(x), \varepsilon)$ be contained in U. Then, there exists a $\delta > 0$ so that $f(B_X(x, \delta)) \subseteq B_Y(f(x), \varepsilon)$ by definition of continuity. This means that $B_X(x, \delta) \subseteq f^{-1}(U)$, i.e. for each $x \in f^{-1}(U)$, we can find an open ball centered at x contained in U, which is what it means for $f^{-1}(U)$ to be open.

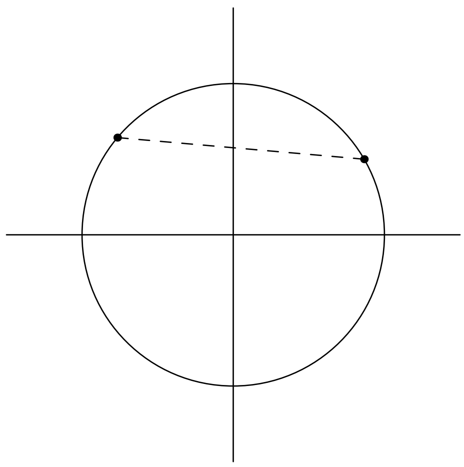
Conversely, suppose $f^{-1}(U)$ is open in X for all open $U \subseteq Y$. Then, let x be in Y and consider the set $f^{-1}(B_Y(f(x), \varepsilon))$. The ball is open in Y, so this pre-image is open in X and contains $f^{-1}(x)$. This means we can find an open ball $B_X(x, \delta) \subseteq f^{-1}(U)$ due to openness, but then the existence of δ is exactly what is required to prove f is continuous, so we are done. □

8.1 Subspaces of metric spaces

Let (X, d) be a metric space. There are a couple of natural ways to generate more metric spaces given a few metric spaces: for example, homework 6 problem 6 showed that the product of two metric spaces is a metric space in a natural way. We can also create a metric space by taking a subset and just restricting the metric to the subset; that is, if $E \subseteq X$, then we can define a metric on E by

$$d_E = d|_{E \times E}.$$

For example, let E be the unit circle $\{(x, y) : x^2 + y^2 = 1\}$. Then, we can define the distance between two points by... well, their distance (in the standard Euclidean sense).



Or, for a simpler toy example, we can take $E = (0, 1] \subseteq \mathbb{R}$. Then, *within* E, an open ball $B_E(z, \varepsilon)$ centered at z with radius ε would be

$$B_E(z, \varepsilon) = \{x \in E : d_E(x, z) < \varepsilon\} = \{x \in X : d(x, z) < \varepsilon\} \cap E = B_X(z, \varepsilon) \cap E.$$

That is, to look at an open ball in E, we take the corresponding open ball in X and chop off the parts outside E (and this generalizes to all open sets). Crucially, this means that

openness is unique to each metric space: a set can be open in E while not being open in X. In our example with $X = \mathbb{R}$, $E = (0, 1]$, $B_E(1, \varepsilon) = (1 - \varepsilon, 1]$ is open in E, but not open in $\mathbb{R}$ (because any open ball in $\mathbb{R}$ centered at 1 would contain numbers bigger than 1).

Exercise 8.2. What do open sets look like on the unit circle, when treated as a subset of $\mathbb{R}^2$ with the usual distance metric?

8.2 Connectedness

Definition 8.3 — A metric space (X, d) is **connected** if there are no nontrivial sets that are both open and closed (clopen). That is, if $A \subset X$ is clopen, then $A = X$ or $A = \varnothing$. We say that $E \subseteq X$ is connected if E is connected with respect to the metric inherited from X.

If the space is not connected then we say it is **disconnected**.

Perhaps the more natural way to think of this is that X is connected if it cannot be expressed as the union of two nonempty disjoint open sets. This is equivalent to the definition because if $X = A \sqcup B$ for open A and B, then $B = X \setminus A = A^c$ and $A = X \setminus B = B^c$ are both complements of open sets and thus open.

Example 8.4

The set $E = (-10, -1) \cup (1, 10)$ with the metric from $\mathbb{R}$ is not connected, as $(-10, 1)$ and $(1, 10)$ are clopen in E.

Exercise 8.5 (Disconnectedness of subspaces)**.** Given metric space (X, d), show that a subset $E \subseteq X$ is disconnected iff there exist disjoint sets $A, B \subseteq X$ that are open in X such that $A \cap E \neq \varnothing$, $B \cap E \neq \varnothing$, $A \cap B \cap E = \varnothing$, and $E \subseteq A \cup B$. Use this to argue that $(-10, -1) \cup (1, 10) \subseteq \mathbb{R}$ is disconnected.

What should the connected sets in $\mathbb{R}$ look like?

Exercise 8.6. Make a guess.

It's not hard to guess that, in fact, the connected sets are just be the intervals.

Theorem 8.7 (Connectedness of open intervals)**.** The open interval $(0, 1)$ is connected.

Of course, this generalizes to any (a, b).

Proof. Suppose that $(0, 1)$ is a disjoint union of two open sets A, B. Take $x \in A$, and define

$$U_x = \{\delta > x : [x, \delta) \subseteq A\} \qquad V_x = \{\delta < x : (x, \delta] \subseteq A\}.$$

That is, U_x tells you "how far away can you go to the right of x before reaching a point outside A?" and V_x encodes the analogous information to the left.

Exercise. Draw a picture!!!

We can see that

$$0 \leq \inf V_x < x < \sup U_x \leq 1$$

(the middle two inequalities are strict by openness of A). However, we cannot have both $\inf V_x = 0$ and $\sup U_x = 1$, since these two together force $A = (0, 1)$.

Exercise. Prove the above statement.

So, suppose WLOG that $\sup U_x < 1$.

Claim 8.7.1 — $y = \sup U_x$ is not in A.

Proof. Suppose $y \in A$. Then, we can find an open ball centered at y, say with radius ε, that is contained in A. Then, $[x, y) \cup (y, y+\varepsilon) = [x, y+\varepsilon) \subseteq A$, so $\sup A > y$, contradiction. □

This means $y \in B$. But then we can find an open ball $(y-\varepsilon, y+\varepsilon) \subseteq B$, so $(y-\varepsilon, y+\varepsilon)$ is disjoint from A. This contradicts the fact that $y = \sup U_x$, since this means that $y - \varepsilon$ is an upper bound for U_x. □

Remark 8.8. The idea behind this proof is that since open sets in $\mathbb{R}$ are unions of open intervals, we want to somehow look at a "boundary" between A and B, as this is where the issue fundamentally happens (just look at how y was an issue regardless of whether it was in A or B!). This is what motivates defining U_x and V_x; we are essentially looking for a maximal open interval (w, y) containing x such that $(w, y) \subseteq A$ (here w would be $\inf V_x$).

The similar result for closed intervals is deferred to the homework, or something.

Theorem 8.9. Let $f\colon X \to Y$ be continuous. Then, if $E \subseteq X$ is connected, $f(E) \subseteq Y$ is connected.

Proof. We leverage Exercise 8.5 to prove the contrapositive. Namely, if $f(E)$ is disconnected, then there exist open subsets A, B of Y that do not intersect inside $f(E)$ and whose union contains $f(E)$. Let $A' = f^{-1}(A)$ and $B' = f^{-1}(B)$, which are both open. Then, A' and B' are disjoint and their union contains $f^{-1}(f(E)) \supseteq E$, which is enough to show that E is disconnected. □

Lemma 8.10. A set $I \subseteq \mathbb{R}$ satisfies the condition that, for all a, $b \in I$,

$$a \leq z \leq b \implies z \in I$$

if and only if it is an interval.

Proof. If I is an interval then the conclusion is trivial.

For the other direction, let $p = \inf I$ and $q = \sup I$. For any ε, we can find p' and q' in I such that

$$p' < p + \varepsilon, \qquad q - \varepsilon < q',$$

so $p + \varepsilon \leq z \leq q - \varepsilon$ implies $z \in I$. Taking ε arbitrarily small gives $p < z < q \implies z \in I$ (where the inequalities are strict because we cannot guarantee that p or q are in I). This is enough to conclude that I is an interval because it is just the set $\{z : p < z < q\}$, potentially with p or q added in. □

Theorem 8.11. $I \subseteq \mathbb{R}$ is connected if and only if it is an interval.

Proof. We have shown (in conjunction with the homework, apparently) that all intervals are connected.

For the other direction, we prove that if I is not an interval then it is disconnected. Lemma 8.10 shows that, since I is not an interval, we can find a and b in I such that there exists a z that is in $[a, b]$ but not I. Then we can take $A = (-\infty, z)$ and $B = (z, \infty)$ and use Exercise 8.5 to show disconnectedness. □

The upshot?

Theorem 8.12. Let (X, d) be a connected metric space such that $f\colon X \to \mathbb{R}$ is continuous. Then, $f(X)$ is an interval.

Proof. $f(X) \subseteq \mathbb{R}$ is connected, so it is an interval by Theorem 8.11. □

Exercise 8.13. Prove Theorem 7.2.

8.3 Path-connectedness

There is a much more intuitive possible way to define connectedness:

Definition 8.14 — We say that $a, b \in X$ are **connected by a path** if there exists a continuous function $\gamma\colon [0, 1] \to X$ such that $\gamma(0) = a$, $\gamma(1) = b$. The space X is **path-connected** if any two points in X are connected by a path.

Theorem 8.15. If X is path-connected, then X is connected.

Sketch. Let X be a union of disjoint open sets A, B. Then, take a path γ between $a \in A$ and $b \in B$ and look at $\gamma^{-1}(A)$, $\gamma^{-1}(B)$. □

Unfortunately the converse of this theorem does not hold.

9 Week 9: 10/22 and 10/24

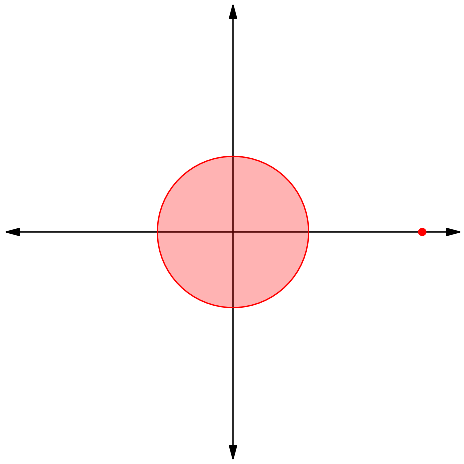
9.1 Accumulation point

Definition 9.1 (Accumulation point) — Let $E \subseteq X$ be a subset of metric space X. We say that $x \in X$ is an **accumulation point** of E if, for every $\varepsilon > 0$, the ball $B_X(x, \varepsilon)$ intersects E at a point **besides** x.

In particular $B_X(x, \varepsilon) \cap E$ must be infinite for any ε in order for x to be an accumulation point. Of course, accumulation points are limit points, since we can find a sequence of points converging to x by making ε smaller and smaller.

The distinction is the following: consider the set

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\} \cup \{(0, 100)\}.$$



(Diagram not to scale.) Then, $(100, 0)$ is a limit point, but not an accumulation point!!

9.2 Basic definition of compactness and Weierstrass's theorem

Last week, we proved the intermediate value theorem via the machinery of connectedness. The other part of the two big theorem in week 7 is Weierstrass's theorem.

Definition 9.2 (Compactness) — Let (X, d) be a metric space. We say that X is compact if, whenever X is written as a union of open sets

$$X = \bigcup_{\alpha \in I} U_\alpha,$$

there exist indices $\alpha_1, \ldots, \alpha_n \in I$ such that

$$X = \bigcup_{i=1}^{n} U_{\alpha_n}.$$

Example 9.3 (Non-example)

Note that $X = \mathbb{R}$ is not compact. To prove this, we need to antagonistically choose the U_α. We can clearly say that

$$\mathbb{R} = \bigcup_{\alpha \in [1, +\infty)} U_\alpha,$$

where $U_\alpha = (-\alpha, \alpha)$. Then, if $\alpha_1, \ldots, \alpha_n$ is a finite set of indices such that WLOG $\alpha_1 \leq \alpha_2 \leq \cdots \leq \alpha_n$, we would get

$$U_{\alpha_1} \cup \cdots \cup U_{\alpha_n} = (-\alpha_n, \alpha_n) \neq \mathbb{R}.$$

In some sense, the point is that we don't have "too much redundancy" with our choice of cover here.

Exercise 9.4. Prove similarly that $(0, 1)$ is not compact.

Some terminology to make all of this easier to talk about:

Definition 9.5 (Covers and subcovers) — We say that a collection of open sets $\mathcal{U} = \{U_\alpha\}$ is a **(open) cover** for E if

$$E \subseteq \bigcup_{U \in \mathcal{U}} U.$$

If $\mathcal{U}$ and $\mathcal{U}'$ are both covers of E such that $\mathcal{U}' \subseteq \mathcal{U}$, then $\mathcal{U}'$ is a **subcover** of $\mathcal{U}$. Obviously, we attach the adjective "finite" to a (sub)cover if the relevant collection of open sets is finite.

So we can more concisely say "X is compact if every cover of X has a finite subcover." As usual, when talking about subsets E of some ambient metric space X, we work with the induced metric, so E is compact if every open cover **in** E has a finite subcover. This turns out to not be an issue at all.

Lemma 9.6. E is compact if and only if every open cover of E **in** X has a finite subcover.

Proof. We will just prove the direction assuming E is compact (in the induced metric); the other direction is left as an exercise. In this case, suppose that $\mathcal{V} = \{V_\alpha\}$ is a cover for E. Then, $U_\alpha = V_\alpha \cap E$ is open in E by Problem 4 in the homework, so $\mathcal{U} = \{U_\alpha\}$ is an open cover for E in E, and then we can extract a finite subcover $\{U_{\alpha_i} : 1 \leq i \leq n\}$ whose union is E, which in turn means we have a finite subcover

$$\bigcup_{i=1}^{n} V_{\alpha_i} \supset \bigcup_{i=1}^{n} U_{\alpha_i} = E.$$

$\square$

Exercise 9.7. Prove the other direction of Lemma 9.6.

This lemma says something very important: when talking about the compactness of a set, we don't need to refer to its parent metric space at all besides to get our open sets! This is very different from, say, deciding when sets are open, which is dependent on whether we are in X or E (for example, $[0, 1)$ and $[0, 2]$ are open in $[0, 2]$ but not $\mathbb{R}$).

Example 9.8 (Compact sets)
Some examples of compact sets:

- All finite sets.
- Closed intervals, as we show in Theorem 9.9.
- Any closed and bounded subset of $\mathbb{R}^n$, in fact — this is called the Heine-Borel theorem.

Theorem 9.9. The interval $[0, 1]$ is compact.

Remark 9.10. In fact, it turns out that the set of rationals q for which $0 \leq q \leq 1$ is not compact! So we really have to make use of the fact that we are living in $\mathbb{R}$ here.

Proof. Let $\mathcal{U} = \{U_\alpha\}$ be an open cover of $[0, 1]$. Now, suppose for the sake of contradiction that we can't find a finite subcover here, and define $I_0 = [0, 1] = [a_0, b_0]$. Then, one of $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ are also covered by $\mathcal{U}$. One of them must also not have a finite subcover, since if both intervals had a finite subcover then we could just merge them together. Let $I_1 = [a_1, b_1]$ be the interval that doesn't have a finite subcover (pick one at random if both are OK).

Then, again, I_1 can be split into $[a_1, \frac{a_1+b_1}{2}]$ and $[\frac{a_1+b_1}{2}, b_1]$, and yet again one of these cannot have a finite subcover, which we will call $I_2 = [a_2, b_2]$.

In this way, we inductively define a sequence of intervals $I_n = [a_n, b_n]$ such that $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$, none of the I_n has a finite subcover in $\mathcal{U}$, and $b_n - a_n = 2^{-n}$.

Exercise 9.11. Prove that (a_n) and (b_n) both converge to the same number x.

The result of the exercise tells us that the intersection of all the I_n is the singleton set

$$\bigcap_{n \geq 0} I_n = [x, x] = \{x\}.$$

We know that x has to belong to some $U \in \mathcal{U}$. But then this means that $B(x, \varepsilon) = (x - \varepsilon, x + \varepsilon) \subseteq U$ for some $\varepsilon > 0$ because U is open. Since a_n and b_n converge to x, both a_n and b_n must belong to this interval eventually, hence $I_n = [a_n, b_n] \subseteq (x - \varepsilon, x + \varepsilon)$ eventually. This means that in fact I_n has a finite cover in $\mathcal{U}$ eventually, so we have found a contradiction! $\square$

Exercise 9.12. Where exactly did we use the fact that $\mathbb{R}$ is complete? Use this to argue that $[0, 1] \cap \mathbb{Q}$ is not compact.

Theorem 9.13 (Preservation of compactness)**.** If $f : X \to Y$ is continuous and $K \subseteq X$ is compact, then $f(K) \subseteq Y$ is also compact.

Exercise 9.14. Prove the theorem above. Also, compare to Theorem 8.9.

Theorem 9.15. Compact subsets are closed.

Proof. Let K be compact subset of metric space X. The goal obviously is to prove that K contains its limit points.

Suppose otherwise, so $\ell \notin K$. Then, consider the open sets

$$U_n = \left\{ x \in X : d(x, \ell) > \frac{1}{n} \right\}.$$

Exercise 9.16. Prove that the U_n really are open in X.

Then the union of the U_n contains every point in X except ℓ, which means the union contains K because of our assumption that ℓ isn't in K. By compactness, we can find some $n_1, \ldots, n_k$ such that

$$K \subseteq U_{n_1} \cup \cdots \cup U_{n_k},$$

but we can see that this union is just U_M where $M = \max\{n_1, \ldots, n_k\}$, which is the complement of a closed ball. That is, there is a ball $B_X(\ell, \varepsilon) \subseteq X \setminus K$ (since the closed ball contains an open ball). The contradiction then comes from taking a sequence (x_n) of points in K that converges to ℓ, as (x_n) must eventually be in $B_X(\ell, \varepsilon)$. $\square$

You are now free to extract the Weierstrass theorem!

Exercise 9.17. Prove Theorem 7.1. You will have to show that $[a, b]$ is the only kind of interval (infinite or not) that is compact.

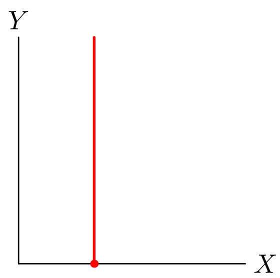
You should also note that the proof that $(0, 1)$ is not compact is completely trivial now due to Theorem 9.15, as opposed to the work that you had to do in Exercise 9.4.

9.3 More properties of compactness

Theorem 9.18. If X and Y are compact, then so is $X \times Y$.

proof

Proof. The idea behind this proof is that if $U \subseteq X \times Y$ is open, then U is a product of sets of the form $B_X(x, \varepsilon) \times B_Y(y, \varepsilon)$.



For some point $x \in X$, we look at the "line" $\{x\} \times Y$. Then, we have a continuous map (actually a homeomorphism) $f_x : Y \to \{x\} \times Y$ given by $y \mapsto (x, y)$, which means this line is compact too. Fixing a cover $\{U_\alpha\}$, we can say that $\{U_{x,i} : 1 \le i \le n(x)\}$ is some finite cover of $\{x\} \times Y$. Each $U_{x,i}$ can be written as $A_{x,i} \cap B_{x,i}$ where $A_{x,i}$ and $B_{x,i}$ are open balls in X and Y respectively. $\square$

finish pf (look at nb)

Corollary 9.19. Given real numbers $a_1, \ldots, a_n, b_1, \ldots, b_n$, the set $[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ is compact.

Another way to generate compact sets:

Lemma 9.20. If X is compact and $K \subseteq X$ is closed, then K is compact.

Proof. Let $\{U_\alpha\}$ be an open cover of K. Then $\{U_\alpha\} \cup \{X \setminus K\}$ is an open cover of X because $X \setminus K = U_0$ is open in X. $\square$

finish

Nicely enough, we can say something much stronger than Corollary 9.19, fully characterizing all compact sets in $\mathbb{R}^n$.

Theorem 9.21 (Heine-Borel)**.** A set $K \subseteq \mathbb{R}^n$ is compact if and only if it is closed in bounded.

When we say that a set E is **bounded**, we mean that the quantity

$$\operatorname{diam}(E) = \sup\{d(x,y) : x, y \in E\}$$

is a real number.

Proof of Theorem 9.21. It is not hard to prove the forward direction.

Lemma 9.22. If K is closed and bounded, it is compact.

Proof. If K is closed and bounded, then $\operatorname{diam}(K) = R < +\infty$, so $K \subseteq B(\mathbf{x}, R)$ where $\mathbf{x}$ is any point in K. We know that the standard metric on $\mathbb{R}^n$ is equivalent to $d_\infty(x,y) = \max_i |x_i - y_i|$, meaning there exists an R' such that $B(\mathbf{x}, R) \subseteq B_\infty(\mathbf{x}, R')$. The balls in this new metric are just products of open intervals, so, $\overline{B_\infty(\mathbf{x}, R')}$ is compact and K is a closed subset of this compact set, thus it is compact. $\square$

The other direction uses very intrinsically the idea that we are working in a metric space (in fact that we are in $\mathbb{R}$).

Definition 9.23 — A metric space is (X,d) is **sequentially compact** if every sequence (x_n) in X has a convergent subsequence (the main point is that **the limit is in** X).

Lemma 9.24. If K is compact, then it is sequentially compact.

Proof. Let $E = (x_n)$ be a sequence of elements in K. If E is finite then we can find a constant subsequence by pigeonhole, so suppose E is infinite.

The goal is to show that E has an accumulation point; suppose otherwise for contradiction. Then, each $z \in E$ is contained by some open set whose intersection with E is just $\{z\}$. Take an open cover U_x indexed by points in X such that, if $z \in E$, then U_z contains no points of E except z (for example, each U_z could be a sufficiently small open ball centered at z). Then on one hand we can find a finite subcover of X by compactness, but on the other hand such a finite subcover would have only finitely many elements of the infinite set E. $\square$

Lemma 9.25. If K is sequentially compact, then it is closed and bounded.

Proof. Suppose that K is not closed. Then, take a limit point ℓ of K that is outside of K, and let (x_n) be a sequence of points in K converging to ℓ; this sequence has no convergent subsequence in K because each subsequence also converges to ℓ.

Suppose now that K is unbounded. Then, for some arbitrary $x \in K$, consider the sets

$$S_n = \{y \in K : n - 1 \le d(x,y) < n\}$$

for positive integers n. These sets form a partition of K. If there existed an N for which $n > N$ implies S_n is empty, then K would be bounded, so infinitely many S_n are nonempty, say $S_{n_1}, S_{n_2}, \ldots$. Then take the sequence (x_{n_k}) such that $x_{n_k} \in S_{n_k}$. Each subsequence is unbounded, but it is easy to show that a convergent sequence must be bounded (exercise!), so (x_{n_k}) has no convergent subsequence.

Therefore K must be both closed and bounded if it is sequentially compact. $\square$

These three lemmas constitute a proof of the theorem! $\square$

9.4 Uniform continuity

Recall the definition of continuity of a function $f\colon X \to Y$: for each $x \in X$, given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

The thing is that δ is dependent on both x and ε. We would like it to be dependent on just ε; that is, we'd like to say for each $\varepsilon > 0$, there exists a universal $\delta > 0$ such that

$$d_X(x, y) < \varepsilon \implies d_Y(f(x), f(y)) < \varepsilon.$$

Theorem 9.26. If the domain of a continuous function $f\colon X \to Y$ is compact, then the function is uniformly continuous.

10 Week 10: 10/29 and 10/31

We begin by proving Theorem 9.26.

Proof of theorem. Suppose f is not uniformly continuous. Then, there exists an $\varepsilon > 0$ such that, for each $\delta > 0$, there are two points $x, y \in X$ such that $d_X(x, y) < \delta$ but $d_Y(f(x), f(y)) \geq \varepsilon$.

In other words, for each $n \in \mathbb{N}$, there exist x_n and y_n such that $d_X(x_n, y_n) < \frac{1}{n}$ and $d(f(x_n), f(y_n)) > \varepsilon$. Now, since X is compact, it is sequentially compact (Lemma 9.24). The trick is that we want convergent subsequences (x_{n_k}) and (y_{n_k}) of (x_n) and (y_n) *with the same indices*, since then the condition on x_n and y_n ensures that the limits of these subsequences are equal.

To generate these convergent subsequences, we do a slick trick: first, let (x_{m_k}) be a convergent subsequence of (x_n). Then, take a convergent subsequence of (y_{m_k}) and call it (y_{n_k}). The sequence (n_k) is a subsequence of (m_k), so (x_{n_k}) is also a subsequence of the convergent sequence (x_{m_k}), thus (x_{n_k}) also converges!

Moreover, we can argue (e.g. by squeeze theorem) that the limits are the same.

> **Remark 10.1.** Can you also say that once you know $x_{m_k} \to \alpha$ you also know $y_{m_k} \to \alpha$ by the distance condition?

But then we have two sequences with the same limit, so $f(x_{n_k})$ and $f(y_{n_k})$ are equal, which contradicts the fact that their distance is at least ε. □

10.1 Completeness

Recall the following definition:

> **Definition 10.2** (Cauchy sequences in $\mathbb{R}$) — A sequence of real numbers (x_n) is **Cauchy** if for each $\varepsilon > 0$, there exists an N such that
> $$m, n > N \implies |x_m - x_n| < \varepsilon.$$

Obviously, we can generalize this to arbitrary metric spaces.

> **Definition 10.3** (Cauchy sequences) — A sequence of points (x_n) in a metric space (X, d) is **Cauchy** if, for each $\varepsilon > 0$, there exists an N such that
> $$m, n > N \implies d(x_m, x_n) < \varepsilon.$$

> **Exercise 10.4.** Prove that all convergent sequences are Cauchy.

Intuitively, the idea is that a sequence is Cauchy if "it gets very bunched up as you go further in the sequence," i.e. it *should* have a limit (even if it doesn't actually converge). This motivates the following definition:

> **Definition 10.5** (Completeness) — (X, d) is **complete** if every Cauchy sequence in X converges in X.

This is one approach to constructing $\mathbb{R}$, actually: we can identify each element of $\mathbb{R}$ with equivalence classes of Cauchy sequences!

> **Lemma 10.6.** Cauchy sequences are bounded.

Proof. The idea is that we can bound cofinitely many of the points of the sequence in a ball of radius ε centered at some x_n; say all x_k for $k > N$ are captured in this ball. Then, it's a matter of growing the ball to have radius bigger than

$$\max(\varepsilon, d(x_n, x_1), d(x_n, x_2), \dots, d(x_n, x_N)),$$

at which point we have captured the whole sequence. □

Theorem 10.7. If X is compact, then it is complete.

Proof. We want to show that any Cauchy sequence in X converges in X. The idea is that if a Cauchy sequence has a convergent subsequence, then it is convergent; indeed, letting ℓ be a subsequential limit, we see that, for any $\varepsilon > 0$,

$$d(x_n, \ell) \leq d(x_{n_k}, x_n) + d(x_{n_k}, \ell) < 2\varepsilon$$

is true eventually. We can now leverage sequential compactness to win instantly. □

Theorem 10.8. $\mathbb{R}^n$ is complete.

Proof. Any Cauchy sequence is bounded, so it can be contained in a closed open ball, which is compact. This lets us see that all Cauchy sequences converge. Yay! □

Theorem 10.9. ℓ^2 is complete.

10.2 Sequences of functions

Definition 10.10 (A first definition) — Let $f_n : I \to \mathbb{R}$ be a sequence of functions and E a subset of I. We say that $f_n \to f$ **pointwise** on E if $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in E$.

examples

Example 10.11
content

We can rewrite the definition as saying that for each $\varepsilon > 0$ and $x \in E$,

$$|f_n(x) - f(x)| < \varepsilon$$

is true eventually, say for $n > N$. Here we run into something similar to the distinction of continuity and uniform continuity: N depends on both x and ε. Indeed, we can define *uniform* convergence:

Definition 10.12 (Uniform convergence) — $f_n \to f$ **uniformly** if the threshold N does not depend on x.

Exercise 10.13. Show that $f_n(x) = \frac{1}{n^2x^2}$ does not converge to $f \equiv 0$ uniformly on $(0, 1)$, while $g_n(x) = \frac{\sin(nx^2)}{n}$ does converge uniformly on $(0, 1)$.

Consider

$$\mathcal{B}(E,\mathbb{R}) = \left\{ f\colon E \to \mathbb{R} : \sup_{x\in E} |f(x)| < +\infty \right\}.$$

Then, we have a metric

$$d(f,g) = \sup_{x\in E} |f(x) - g(x)|.$$

Under this metric, $d(f,g) = \varepsilon$ means that at each $x \in E$, f and g are at most ε apart. This is nice because it encodes uniform convergence as follows:

Lemma 10.14. $f_n \to f$ uniformly on E iff $f_n \to f$ in $\mathcal{B}(E,\mathbb{R})$.

Theorem 10.15. $\mathcal{B}(E,\mathbb{R})$ is complete.

Proof. Consider a Cauchy sequence of functions $f_n(x)$. Then, the sequence of numbers $f_n(c)$ is Cauchy for all $c \in E$, so f_n has a pointwise limit f. This is a bounded function: for $x \in E$, we can pick N so that $n > N$ implies $|f_n(x) - f(x)| < 1$ (by definition of convergence), so, for $n > N$,

$$\begin{aligned} |f(x)| &\le |f_n(x)| + |f_n(x) - f(x)| \\ &\le \sup_{x\in E} |f_n(x)| + 1 < +\infty. \end{aligned}$$

To show that this converges with respect to our actual metric d, pick an $\varepsilon > 0$ and take m and n to be integers that are large enough that $d(f_n, f_m) < \varepsilon$; then,

$$\begin{aligned} |f_n(x) - f(x)| &\le |f_n(x) - f_m(x)| + |f_m(x) - f(x)| \\ &\le d(f_n, f_m) + |f_m(x) - f(x)| \\ \implies |f_n(x) - f(x)| &< \varepsilon + |f_m(x) - f(x)|. \end{aligned}$$

By pointwise convergence we can take the limit as $m \to \infty$ and take the supremum to get that $\sup_{x\in E} |f_n(x) - f(x)| = d(f_n, f) \le \varepsilon$ for all $\varepsilon > 0$. $\square$

Definition 10.16 — We call $C^0(E,\mathbb{R})$ the space of continuous functions mapping $E \to \mathbb{R}$, and $C^0_b(E,\mathbb{R})$ the space of bounded continuous functions from E to $\mathbb{R}$.

Theorem 10.17. $C^0_b(E,\mathbb{R})$ is a closed subset of $\mathcal{B}(E,\mathbb{R})$.

Proof. Let f_n be a sequence of continuous bounded functions that converge to some $f \in \mathcal{B}(E,\mathbb{R})$. We wish to show that f is continuous; let $\varepsilon > 0$. By triangle inequality

$$|f(x) - f(x_0)| \le |f(x_0) - f_n(x_0)| + |f_n(x_0) - f_n(x)| + |f_n(x) - f(x)|.$$

Take n large so that the first and third terms are smaller than $\varepsilon/3$ (since $f_n \to f$ uniformly — we bound both by $d(f, f_n)$), and then, since f_n continuous, take $|x_0 - x| < \delta$ so that the middle term is smaller than $\varepsilon/3$.

Remark 10.18. The bound of $|f_n(x) - f(x)|$ cannot be done with just pointwise convergence because then δ would have to depend on n.

Thus $|f(x) - f(x_0)| < \varepsilon$ for $|x - x_0| < \delta$, so f is continuous. $\square$

Corollary 10.19. $C_b^0(E, \mathbb{R})$ is complete.

Corollary 10.20. If E is compact, then $C^0(E, \mathbb{R})$ is complete.

Proof. ...because $C^0(E, \mathbb{R}) = C_b^0(E, \mathbb{R})$ if E is compact. □

The magic of this theory is that we can create entirely new continuous functions as limits of sequences of pretty simple functions. The standard example of this is **power series**: take real numbers c_0, c_1, c_2, ..., and define

$$f(x) := c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots = \lim_{n \to \infty} \sum_{k=0}^{n} c_k x^k.$$

In other words f is the limit of functions $f_n(x) = c_0 + c_1 x + \cdots + c_n x^n$. The question, then, is when the limit exists. For this, the root test wants us to find

$$\alpha = \limsup |c_k x^k|^{1/k} = |x| \cdot \limsup |c_k|^{1/k} := |x| \cdot \beta.$$

Root test thus tells us that the limit exists when $|x| < \frac{1}{\beta}$ and doesn't exist when $|x| > \frac{1}{\beta}$, so really $\frac{1}{\beta}$ is the thing we care about — we will call it R, the **radius of convergence**. To clarify: f is defined as a pointwise limit when $x \in (-R, R)$, it might be defined at $x = \pm R$ (since root test is inconclusive), and it is never defined for $|x| > R$.

Example 10.21 (Exponential function)

The exponential function exp can be defined as the power series

$$\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

It is easy to check that its radius of convergence is all of $\mathbb{R}$. The familiar properties of the exponential function are also not too hard to check. We define Euler's number e to then be $\exp(1)$.

In view of the theory above, we'd really like this convergence to be uniform on $(-R, R)$.

Theorem 10.22 (Weierstrass M-test). Let $E \subseteq \mathbb{R}$. Suppose that (g_n) is a sequence of (bounded) functions $E \to \mathbb{R}$ such that $|g_k(x)| < M_k$ for all $x \in E$. Then, if $\sum M_k$ converges, $f_n = \sum_{k=1}^{n} g_n$ converges uniformly.

Proof Sketch. Prove that f_n is Cauchy by using the fact that the sequence of partial sums of M_k is Cauchy. □

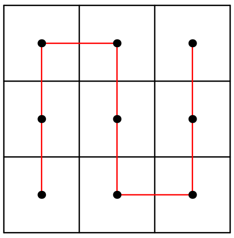
Exercise 10.23. Show that a power series converges uniformly on every compact subset of its interval of (pointwise) convergence. Show also that we can't improve this to uniform convergence on $(-R, R)$.

10.3 A really cool application

Theorem 10.24 (Space filling curve). There exists a function $f\colon [0,1] \to [0,1]^2$ that is both continuous and surjective.

Exercise 10.25. Show that f cannot *also* be injective — equivalently, prove that $[0,1]$ and $[0,1]^2$ are not homeomorphic.

Proof. Split the square into nine smaller squares of equal size and draw a path through their centers as follows:



Then, for any partition into 9^k squares, we can patch these mini-pieces together as follows:

param

Parameterize this path by [placeholder] to get maps $f_n\colon [0,1] \to [0,1]^2$.

Claim 10.25.1 — For all m and n,

$$|f_n(x) - f_m(x)| \le \sqrt{2} \cdot 3^{-\min(m,n)}.$$

pf of claim

Proof. todo □

Then, the limit of the f_n is a surjective continuous function! □

11 Week 11: 11/5 and 11/7

11.1 Limits of functions

Definition 11.1 (Limits of functions) — Let E be a subset of X and consider $f\colon X \to Y$. Also, let $x_0 \in X$ be an accumulation point of E. Then,

$$\lim_{x \to x_0} f(x) = L$$

means that for each sequence (x_n) of points in E converging to x_0 such that $x_n \neq x_0$ for all $n \in \mathbb{N}$, we have $f(x_n) \to L$. To emphasize: the limit does not care about $f(x_0)$.

The expected limit uniqueness and combination properties hold.

Theorem 11.2. If $f\colon \mathbb{R} \to \mathbb{R}$ is continuous and

$$\lim_{x \to \infty} f(x) = L_1, \qquad \lim_{x \to -\infty} f(x) = L_2$$

for real L_1 and L_2, then f is bounded.

Proof. Suppose for the sake of contradiction that (WLOG)

$$\sup_{x \in \mathbb{R}} f(x) = \infty.$$

Then, we can find a sequence of real numbers (x_n) such that the sequence $(f(x_n))$ diverges to ∞.

Claim 11.2.1 — (x_n) is unbounded.

Proof. If (x_n) were bounded, then we can find a convergent subsequence (x_{n_k}) with limit ℓ. By continuity, the limit of $(f(x_{n_k}))$ is $f(\ell) \in \mathbb{R}$, but then $(f(x_n))$ has a limit, so the limit must be $f(\ell)$, contradicting the fact that the limit is ∞. □

But now we have found a sequence such that $x_n \to \infty$ or $x_n \to -\infty$ such that $f(x_n)$ blows up to ∞, which contradicts the fact that $f(x_n)$ should converge to either L_1 or L_2. □

Definition 11.3 (Limits of functions, again) — Let (X, d_X) and (Y, d_Y) be metric spaces and let $f\colon X \to Y$ be a function. Then,

$$\lim_{x \to x_0} f(x) = \ell$$

is equivalent to saying that, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < d_X(x, x_0) < \delta \implies d_Y(\ell, f(x)) < \varepsilon.$$

11.2 Derivatives

And now comes the part where we actually have to live in $\mathbb{R}$.

Definition 11.4 (Differentiability) — Let $I \subseteq \mathbb{R}$ be an open interval. A function $f\colon I \to \mathbb{R}$ is **differentiable** at a point $x_0 \in I$ if the limit

$$f'(x_0) := \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists in $\mathbb{R}$. We call this limit the **derivative** of f at x_0.

Exercise 11.5 (Equivalent formulation)**.** Show that if f is differentiable at x_0, then

$$f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h}.$$

Lemma 11.6 (Differentiable implies continuous)**.** If $f\colon I \to \mathbb{R}$ is differentiable at x_0, then it is continuous at x_0.

Proof. By limit properties,

$$\lim_{x\to x_0} f(x) - f(x_0) = \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0} \cdot (x-x_0) = f'(x_0)\cdot 0 = 0,$$

so $f(x) \to f(x_0)$ as $x \to x_0$. (Or you can get into the details with sequences to prove the same thing.) $\square$

Theorem 11.7 (Derivative is zero at extrema)**.** If $f\colon I \to \mathbb{R}$ is differentiable at x_0 and x_0 is a local maximum (minimum) of f, i.e. $f(x) \leq f(x_0)$ ($f(x) \geq f(x_0)$) for all x in some open ball containing x, then $f'(x_0) = 0$.

Proof. We prove the theorem when x_0 is a local maximum, i.e. $f(x_0) \geq f(x)$ whenever $|x - x_0| < \varepsilon$. Suppose for the sake of contradiction that $f'(x_0) \neq 0$, and say WLOG that the derivative is positive. Then, there is a $\delta > 0$ such that

$$\frac{f(x)-f(x_0)}{x-x_0} > 0$$

whenever $|x - x_0| < \delta$. Picking x so that $0 < x - x_0 < \min(\delta, \varepsilon)$, this forces $f(x) - f(x_0) > 0 \iff f(x) > f(x_0)$, contradiction. $\square$

Theorem 11.8 (Rolle's theorem)**.** Let $f\colon [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b). If $f(a) = f(b)$, then there exists $x \in (a, b)$ for which $f'(x) = 0$.

Proof. By Weierstrass's theorem, there exist points m and M such that

$$f(m) \leq f(x) \leq f(M)$$

for all $x \in [a, b]$. If f is constant then the result is trivial (the derivative is zero everywhere), so suppose f isn't constant. Then, say WLOG $f(M) \neq f(a)$. Since M is a local maximum of f, Theorem 11.7 says that $f'(M)$ is zero. $\square$

Theorem 11.9 (Mean value theorem)**.** If $f\colon [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b), then there exists an $x \in (a, b)$ such that

$$\frac{f(b)-f(a)}{b-a} = f'(x).$$

Exercise 11.10. Prove Theorem 11.9. (You'll have to use Rolle's theorem.)

Corollary 11.11. Prove that if $f\colon (a,b) \to \mathbb{R}$ is a differentiable function then $f'(x) = 0$ for all x iff f is a constant function.

Exercise 11.12. Prove Corollary 11.11. Prove that if $f, g\colon (a,b) \to \mathbb{R}$ are differentiable then $f'(x) = g'(x)$ for all x iff $f - g$ is a constant function.

Corollary 11.13. Let $f\colon (a,b) \to \mathbb{R}$ be differentiable.

(i) If $f' \geq 0$ then f is increasing.

(ii) If $f' \leq 0$ then f is decreasing.

(iii) If the inequalities are strict in (i) and (ii) then the monotonicity is also strict.

Proof. We just prove that $f' > 0$ implies f strictly increasing. This means we want to prove that $y > x$ implies $f(y) > f(x)$. Indeed, let $y > x$. There exists a $\xi \in (x,y)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(\xi) > 0,$$

so $y - x > 0$ implies $f(y) - f(x) > 0$ too. □

11.3 Sequences of functions, again

Definition 11.14 (C^k functions) — A function is said to be **class C^k** if it is differentiable k times, and all derivatives (in particular the kth derivative) are continuous. So, for example, C^0 functions are just the continuous ones and C^1 functions have continuous derivative.

Remark 11.15. We don't really care about functions with discontinuous derivative.

Now, as we did last week, we'd like to look at functions as a whole. We saw as a corollary of 10.17 that $C^0[a,b]$ (the continuous functions $[a,b] \to \mathbb{R}$) form a complete metric space. We know that $C^1[a,b] \subseteq C^0[a,b]$, and we'd hope that, like $C^0[a,b]$, that this is a complete space. We are not so lucky.

- There exist sequences (f_n) of C^1 functions with limit f not in C^1.
- There exist sequences (f_n) of C^1 function with limit f in C^1... but f_n' need not converge to f.

So what do we do?? The idea is to make our metric reflect what we want to control.

Definition 11.16 (C^1 metric) — Let $d_{C^0} := d$ be the familiar metric on C^0. We define a new metric

$$d_{C^1}(f,g) = d_{C^0}(u,v) + d_{C^0}(u',v').$$

Of course, we can generalize this to C^k as

$$d_{C^k}(f,g) = d_{C^0}(u,v) + d_{C^{k-1}}(u',v') = \sum_{i=0}^{k} d_{C^0}(u^{(i)}, v^{(i)}).$$

Theorem 11.17. With this new metric, $C^1[a,b]$ is complete.

Proof. Given a Cauchy sequence (u_n) in $(C^1[a,b], d_{C^1})$, the point is that both (u_n) and (u_n') are Cauchy with respect to d_{C^0}. If $u_n \to u$ and $u_n' \to v$, we wish to show that u is class C^1 and $u' = v$.

Fix $x \in [a,b]$. Then, consider the functions

$$\varphi_n(t) = \frac{u_n(t) - u_n(x)}{t - x}, \qquad \varphi(t) = \frac{u(t) - u(t)}{t - x}.$$

We know that

$$\lim_{n\to\infty} \varphi_n(t) = \varphi(t)$$
$$\lim_{t\to x} \varphi_n(t) = u_n'(t)$$
$$\lim_{t\to x} \varphi(t) = u'(t)$$

wait why does u' exist

pointwise.

Claim 11.17.1 — $\varphi_n \to \varphi$ uniformly.

todo

Proof. todo □

Claim 11.17.2 — If $\varphi_n \to \varphi$ uniformly on some E and x is an accumulation point, then

$$\lim_{n\to\infty} \lim_{t\to x} \varphi_n(t) = \lim_{t\to x} \lim_{n\to\infty} \varphi_n(t),$$

i.e. we can swap the limits if the convergence of φ is uniform.

Proof. Let

$$\lim_{t\to x} \varphi_n(t) = \ell_n$$

and suppose that $\ell_n \to \ell$, i.e. the LHS is ℓ. Then, the inner limit on the right is $\varphi(t)$, so

$$|\ell - \varphi(t)| \le |\varphi(t) - \varphi_n(t)| + |\varphi_n(t) - \ell_n| + |\ell_n - \ell|$$

what?

and each piece can be bounded independently of ε as long as $|t - x|$ is small enough. □

□

Exercise 11.18. Generalize to $C^k[a,b]$.

13 Week 13: 11/19 and 11/21

13.1 Derivatives of power series

We're going to use the theory from the latter portion of Week 11 to look at derivatives of power series. The situation is that we have a power series f given by the limit of

$$f_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

on the interval $(-R, R)$ of convergence. Restricting to the interval $[-R', R'] \subseteq (-R, R)$, we look at the sequence

$$f_n'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots + na_nx^{n-1}.$$

Both f_n and f_n' converge uniformly on $[-R', R']$, so f_n converges in $C^1[-R', R']$.

> **Exercise 13.1.** Show that the radius of convergence of f_n' is the same as that of f_n.

Via the theory in Week 11, we can see that the derivative of f is just the power series we get by differentiating termwise.

You should then be able to easily conclude the following:

Theorem 13.2. Power series are infinitely differentiable on their interval of convergence!

Now, we play!

Example 13.3 (Geometric series)
We know that $f(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. Then,

$$f'(x) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1},$$

and you can spam derivatives to get more and more facts like this.

Example 13.4 (Taylor series)
Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots$ be a power series. Then, we have

$$f'(x) = 1 \cdot a_1 + 2a_2x + 3a_3x^2 + \cdots,$$

which, evaluated at 0, gives $1 \cdot a_1$. Similarly

$$f''(0) = 2 \cdot 1 \cdot a_2, \quad f^{(3)}(0) = 3 \cdot 2 \cdot 1 \cdot a_3,$$

and in general $f^{(n)}(0) = n! \cdot a_n$. Thus, we can write

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

In particular,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n.$$

The caveat is that

$$\limsup \left| \frac{f^{(n)}(0)}{n!} \right|^{1/n}$$

need not behave nicely.

Exercise 13.5. Find an example where the Taylor series has a radius of convergence of 0.

Lemma 13.6. Let $f\colon (a,b) \to \mathbb{R}$ and let $c \in (a,b)$. Suppose also that $f \in C^n(a,b)$. Define

$$P_k(x) = \sum_{j=0}^{k} \frac{f^{(j)}(c)}{j!} \cdot (x-c)^j,$$

and let

$$R_n(x) = f(x) - P_{n-1}(x).$$

Then, for a fixed x, there exists a y between c and x such that

$$R_n(x) = \frac{f^{(n)}(y)}{n!} \cdot (x-c)^n.$$

Proof. This should be screaming MVT; we need to find a suitable function to use MVT on. We can write

$$f(x) - P_{n-1}(x) = \frac{M \cdot (x-c)^n}{n!}$$

for some constant M. Then, consider

$$g(t) = P_{n-1}(t) + \frac{M \cdot (t-c)^n}{n!} - f(t).$$

In fact, all derivatives (up to the nth) of g are zero at c and some point between c and x, so we can look at the nth derivative and win ☐

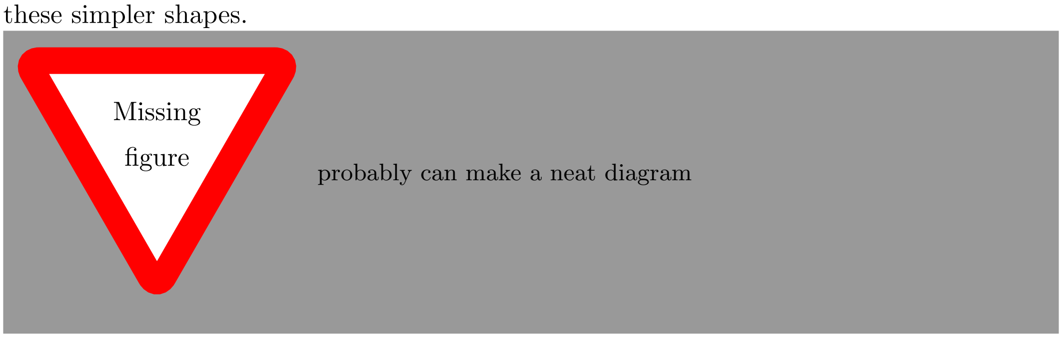
write up thoroughly

13.2 Integration!!!!

Warning — I was half asleep on the days that the integration lessons were taught, so these notes are a post-lesson rendition based primarily on my own knowledge. As such, some of the notation and content here may not align with the stuff shown in class.

As we all know from high school, the goal of integration is to make the idea of "area under a graph" more precise.

In high school geometry, we learned how to compute areas of simple shapes, like rectangles and triangles; to compute areas of more complicated shapes, we try to decompose them into these simpler shapes.



However, this is hard (impossible) for fucky shapes like circles, so we resort to approximations from inside and outside.

approximating circles w ngons

Let's start with extra basic functions. The most basic function we can come up with is

$$\varphi_{(a,b)}(x) = \begin{cases} 1 & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

i.e. the indicator function of the interval (a, b). More generally, for a (finite) interval I, we write

$$\varphi_I(x) = \begin{cases} 1 & x \in I \\ 0 & \text{otherwise} \end{cases}$$

We call these **characteristic** or **indicator** functions. (In class, we also use **simple** functions, but I don't like that terminology.)

The area under the graph of $\varphi_{(a,b)}$ should obviously be $b - a$, so we write

$$\int_{-\infty}^{\infty} \varphi := \ell((a, b)) = b - a.$$

Definition 13.7 (Step function) — Consider some finite set of intervals $I_1, I_2, \ldots, I_n$ and some real constants $c_1, c_2, \ldots, c_n$. We call the function

$$\varphi := \sum_{k=1}^{n} c_k \varphi_{I_k},$$

a **step function**.

Exercise 13.8. Show that step functions form a vector space. That is, show that (finite) linear combinations of step functions are also step functions.

We would like to define the integral of φ in the obvious way, namely

$$\int_{-\infty}^{\infty} \varphi = \sum_{k=1}^{n} c_k \int_{-\infty}^{\infty} \varphi_{I_k}.$$

In order for this definition to be sound, we have to check the following:

Exercise 13.9. Let φ be a step function, with value c_k on the interval I_k for $1 \le k \le n$. Then, the expression

$$\sum_{k=1}^{n} c_k \int_{-\infty}^{\infty} \varphi_{I_k}$$

is invariant of the choice of I_ks.

(Hint: given two sets of intervals $I_1, \ldots, I_n$ and $J_1, \ldots, J_m$, consider a "common refinement" of the intervals such that each I_a and J_b can be written as a disjoint union of the intervals in the refinement.)

This behaves as we would expect the integral to behave. Specifically:

Exercise 13.10 (Step function integral properties)**.** Show that the integral is a linear function on the set of step functions. That is,

$$\int_{-\infty}^{\infty} c_1\varphi_1 + c_2\varphi_2 = c_1 \int_{-\infty}^{\infty} \varphi_1 + c_2 \int_{-\infty}^{\infty} \varphi_2$$

for any step functions φ_1, φ_2 and real numbers c_1, c_2. Show also that if $\varphi_1 \geq \varphi_2$ on all of $\mathbb{R}$ then their integrals obey the same inequality.

We now have a large enough set of functions that we can start doing approximation.

Definition 13.11 (Darboux integral) — Let $f\colon \mathbb{R} \to \mathbb{R}$ be a bounded function with compact support (i.e. the subset of $\mathbb{R}$ on which $f(x) \neq 0$ can be contained in some compact set). Then, let S_{top} be the set of step functions φ for which $\varphi \geq f$ on all of $\mathbb{R}$ and S_{bot} be the set of step functions φ for which $\varphi \leq f$ on all of $\mathbb{R}$. We define the **upper Darboux integral** of f to be

$$\overline{\int_{-\infty}^{\infty}} f := \inf \left\{ \int_{-\infty}^{\infty} \varphi : \varphi \in S_{top} \right\}$$

and the **lower Darboux integral** of f to be

$$\underline{\int_{-\infty}^{\infty}} f := \sup \left\{ \int_{-\infty}^{\infty} \varphi : \varphi \in S_{bot} \right\}.$$

If these two values coincide, then we can define the **Darboux integral** of f as the common value of its upper and lower Darboux integrals.

Remark 13.12. In your calculus class, you probably saw Riemann integration, where you integrate over an interval $[a, b]$ by picking some partition of the interval $a = x_0 \leq x_1 \leq \cdots \leq x_n = b$ and then picking arbitrary values $x_{k-1} \leq x_k^* \leq x_k$ so that you can estimate your function on $[x_{k-1}, x_k]$ with the constant value $f(x_k^*)$. Then, we can compute the integral of this approximation and take the limit as the subintervals of the partition become arbitrarily small. This turns out to be equivalent to integration in the sense of Darboux, so we need not worry about using "Darboux integrable" and "Riemann integrable" interchangeably.

Exercise 13.13. Prove that the integral properties from Exercise 13.10 extend to the Darboux integral.

Exercise 13.14. Show that unbounded functions are not Darboux integrable.

The following lemma makes it much easier to show whether a function is integrable.

Lemma 13.15 (Integrability criterion). Let $f\colon \mathbb{R} \to \mathbb{R}$ be a bounded function with compact support. Then, f is integrable if and only if, for each $\varepsilon > 0$, there exist step functions $\varphi_1 \le f \le \varphi_2$ such that

$$\int_{-\infty}^{\infty} \varphi_2 - \varphi_1 \le \varepsilon.$$

Proof. If f is integrable, then there is a step function φ_1 bounded above by f that is at most $\varepsilon/2$ away from the integral of f and a step function φ_2 bounded below by f that is at most $\varepsilon/2$ away from the integral of f, by definition of sup/inf; thus the integral of the difference between φ_1 and φ_2 is at most ε.

Conversely, if we can always find step functions like described, then, for each n, then the idea is to construct an increasing sequence of step functions below f and a decreasing sequence of step functions above f and then conclude that the limit of the increasing sequence and decreasing sequence of integrals are the same. $\square$

details

Now, we can finally show that a fairly large class of functions is integrable. First, let's define integration on an interval.

Definition 13.16 — The integral of a bounded function $f\colon \mathbb{R} \to \mathbb{R}$ on an interval $[a, b]$ is equal to

$$\int_a^b f := \int_{-\infty}^{\infty} f \cdot \varphi_{[a,b]},$$

recalling that φ_I is the indicator function of the interval I.

Theorem 13.17 (Increasing functions are integrable). Let $f\colon \mathbb{R} \to \mathbb{R}$ be increasing. Then f is integrable on any interval $[a, b]$.

Theorem 13.18 (Continuous functions are integrable). Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous. Then f is integrable on any interval $[a, b]$.

Proof. Pick $\varepsilon > 0$. Then, since $[a, b]$ is compact, f is uniformly continuous on $[a, b]$. Thus there exists a $\delta > 0$ such that we can find $a = x_0 < x_1 < \cdots < x_n = b$ satisfying $x_i - x_{i-1} < \delta$ for $1 \le i \le n$ and the minimum m_i of f over the interval $[x_{i-1}, x_i]$ differs from the maximum M_i by at most $\frac{\varepsilon}{b-a}$. If we take the step function ϕ that is m_i on $[x_{i-1}, x_i]$ and the step function ψ that is M_i on $[x_{i-1}, x_i]$, then $\phi \le f \le \psi$ on $[a, b]$ and

$$\int_a^b \psi - \phi = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) \le \varepsilon \sum_{i=1}^{n} x_i - x_{i-1} = \frac{\varepsilon}{b-a} \cdot (b - a) = \varepsilon.$$

We are then done by Lemma 13.15. $\square$

We can also finally prove the fundamental theorem of calculus.

Lemma 13.19 (Mean value theorem for integrals). Let $f\colon [a, b] \to \mathbb{R}$ be a continuous function. Then, there is some $c \in [a, b]$ for which

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

Proof. Bound the integral below by $m := \min_{x\in[a,b]} f(x)$ and above by $M := \max_{x\in[a,b]} f(x)$. The existence of c is then guaranteed by the intermediate value theorem. □

Theorem 13.20 (First fundamental theorem of calculus)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function. For a fixed $a \in \mathbb{R}$, define

$$F(x) := \int_a^x f(t)\, dt.$$

Then, F is differentiable with $F'(x) = f(x)$ for all $x \in \mathbb{R}$.

Proof Sketch. Fix some $x \in \mathbb{R}$. Then, if $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\, dt.$$

By Lemma 13.19, the right-hand side is equal to $f(\alpha_h)$ for some $\alpha_h \in [x, x+h]$. As $h \to 0^+$, the α_h approach x, so $f(\alpha_h)$ goes to $f(\alpha)$. We can get an analogous result for when $h < 0$, so the limit of the difference quotient ends up just being $f(x)$. □

15 Week 15

Theorem 15.1 (Second fundamental theorem of caluclus)**.** If $G\colon [a,b] \to \mathbb{R}$ is such that $G'(x) = f(x)$ for all $x \in [a,b]$ and f is Riemann integrable on $[a,b]$, then

$$\int_a^b f(t)\, dt = G(b) - G(a)$$

Proof. Consider $F(x) = \int_a^x f(t)\, dt$ and set

$$H(x) := F(x) - G(x).$$

Claim 15.1.1 — H is constant on $[a,b]$.

Proof. We can compute

$$H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0$$

for all $x \in [a,b]$. □

wait why is F dif'able? we don't know that f is cont

Then, we get

$$F(a) - G(a) = H(a) = H(b) = F(b) - G(b),$$

or equivalently $F(b) - F(a) = G(b) - G(a)$, and then

$$G(b) - G(a) = F(b) - F(a) = \int_a^b f(t)\, dt - \int_a^a f(t)\, dt = \int_a^b f(t)\, dt$$

as desired. □

Recall that $\mathcal{B}([a,b], \mathbb{R})$ is the space of bounded functions $[a,b] \to \mathbb{R}$, which is a complete metric space under the standard sup metric.

Consider the subset $X \subseteq \mathcal{B}([a,b], \mathbb{R})$ consisting of the space of integrable functions.

Theorem 15.2. X is closed (hence complete).

Proof. Let f_n be a sequence of functions in X converging uniformly to some f. We can bound

$$|f - f_n| \leq \frac{\varepsilon}{2(b-a)}$$

for large enough n, independently of x. Then, upper and lower Darboux integrals are monotonic, so

$$\underline{\int_a^b} f_n(x) - \frac{\varepsilon}{2(b-a)}\, dx \leq \underline{\int_a^b} f(x)\, dx \leq \overline{\int_a^b} f(x)\, dx \leq \overline{\int_a^b} f_n(x) + \frac{\varepsilon}{2(b-a)}\, dx.$$

Thus the difference between the upper and lower integrals of f over $[a, b]$ is bounded by ε (since the upper and lower integrals of f_n are equal by hypothesis), i.e. the upper and lower integrals coincide, so uniform limits of Riemann integrable functions are still Riemann integrable. □

Corollary 15.3. Let f be a power series with interval of convergence $(-R, R)$. Then f is integrable on every compact interval contained in $(-R, R)$.

Proof. Power series converge uniformly on compact subsets of their interval of convergence, so we can appeal to Theorem 15.2. □

Consider now the map $T\colon X \to \mathbb{R}$ given by $T(u) = \int_a^b f(t)\, dt$.

Theorem 15.4. T is a Lipschitz function (hence uniformly continuous).

Sketch. Just write out the definition of Lipschitz function; use linearity and triangle inequality to bound

$$|T(u) - T(v)| \leq (b-a)d(u, v).$$ □

For integrals of functions that are bounded but not compactly supported, we define

$$\int_0^\infty f(x)\, dx := \lim_{a\to\infty} \int_0^a f(x)\, dx$$

and similarly for integrals from $-\infty$ to 0, and then we glue them together for something like

$$\int_{-\infty}^\infty f(x)\, dx := \lim_{T\to\infty} \int_0^T f(x)\, dx + \int_{-T}^0 f(x)\, dx.$$

15.1 Differential equations

Let $\Omega \subseteq \mathbb{R}^2$ be open (for simplicity, let $\Omega = I \times \mathbb{R}$ for some open interval I) and consider a function $g\colon \Omega \to \mathbb{R}$ that is continuous. We would like to look for functions $u\colon \mathbb{R} \to \mathbb{R}$ such that

$$u'(t) = g(t, u(t))$$

with a prescribed value $u(t_0) := u_0$. For example, we can have $g(t, u) := \sqrt{t^2 + u}$ so that we want

$$u'(t) = \sqrt{t^2 + u(t)}$$

and we can prescribe $u(0) = 1$.

In many cases, we at least would like to know that a solution u exists, even if it isn't actually tractable.

The game plan is as follows. Suppose for a moment that we have a solution u. Then, we can integrate so get

$$\int_0^x g(t, u(t))\, dt = \int_{t_0}^x u'(t)\, dt = u(x) - u(t_0) = u(x) - u_0,$$

so we get the rearrangement

$$u(x) = u_0 + \int_{t_0}^x g(t, u(t))\, dt.$$

We know a priori that u is differentible, hence continuous, so the integral in this expression above is a class C^1 function. We can thus deduce that if a function solves this equation then it must be class C^1. (In general if g is class C^k then we can deduce that u is class C^{k+1}).

Theorem 15.5. There exists a $\delta > 0$ and a C^1 function $u\colon (-\delta + t_0, \delta + t_0) \to \mathbb{R}$ that satisfy the constraints we want (namely $u'(t) = g(t, u(t))$ and $u(t_0) = u_0$), if g is C^1.

Proof. Let X_δ be the set of continuous bounded functions $u\colon (-\delta + t_0, \delta + t_0) \to \mathbb{R}$, which we know is complete. Consider the operator $T\colon X_\delta \to X_\delta$ defined as

$$u \mapsto Tu(x) := u_0 + \int_{t_0}^x g(t, u(t))\, dt.$$

We will show that

$$\boxed{\text{The operator } T \text{ has a fixed point.}}$$

This will be done via the following theorem.

Theorem (Banach-Cacciopolli fixed point theorem)**.** Let (X, d) be a complete metric space and suppose that $T\colon X \to X$ is a **contraction**, meaning that it is Lipschitz and that its Lipschitz constant L is less than 1. Then, T has a *unique* fixed point.

Proof. Let u_0 be a point in X and consider the sequence $(u_n) = (u, Tu, T^2u, \dots)$ given by $u_0 := u$ and $u_{n+1} = Tu_n$ for $n \geq 0$. Then,

$$\begin{aligned} d(u_n, u_{n+1}) &= d(Tu_{n-1}, Tu_n) \\ &\leq Ld(u_{n-1}, u_n) \\ &\leq L^2 d(u_{n-2}, u_{n-1}) \\ &\vdots \\ &\leq L^n d(u_0, u_1). \end{aligned}$$

Defining $C := d(u_0, u_1)$, we have that $d(u_n, u_{n+1}) \leq CL^n$ for $L < 1$. This lets us estimate $d(u_n, u_m)$ from above by $d(u_n, u_{n+1}) + d(u_{n+1}, u_{n+1}) + \dots$ so that

$$d(u_n, u_m) \leq C \cdot (L^n + L^{n+1} + \cdots) = \frac{CL^n}{1 - L},$$

and we get the same bound with m, so as long as $\min(m, n)$ is large enough, the distance between the terms is going to be smaller than any given $\varepsilon > 0$. Thus the u_n are Cauchy, so they converge (by completeness).

This means

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} T(u_{n-1}) = T(\lim_{n\to\infty} u_{n-1}) = T(\lim_{n\to\infty} u_n),$$

so the limit of the u_n is a fixed point. To show uniqueness, we can see that if u and v are two distinct fixed points of T, then

$$d(u,v) = d(Tu,Tv) \leq L \cdot d(u,v) < d(u,v),$$

which is a contradiction. $\square$

To finish, we claim that

Claim 15.5.1 — The operator $T\colon X_\delta \to X_\delta$ is Lipschitz.

Proof. For two functions u, v,

$$|Tu(x) - Tv(x)| = \left| \int_{t_0}^{x} g(t,u(t)) - g(t,v_t)\ dt \right| \leq \int_{t_0}^{x} |g(t,u(t)) - g(t,v(t))|\ dt.$$

proof

$\square$

$\square$